

Translated

From: e-maxx.ru

Finding Power of Factorial Divisor

You are given two numbers n and k . Find the largest power of k such that $n!$ is divisible by k^x .

Prime k

Let's first consider the case of prime k . The explicit expression for factorial

$$n! = 1 \cdot 2 \cdot 3 \cdot \dots \cdot (n-1) \cdot n$$

Note that every k -th element of the product is divisible by k , i.e. adds $+1$ to the answer; the number of such elements is $\lfloor \frac{n}{k} \rfloor$.

Next, every k^2 -th element is divisible by k^2 , i.e. adds another $+1$ to the answer (the first power of k has already been counted in the previous paragraph). The number of such elements is $\lfloor \frac{n}{k^2} \rfloor$.

And so on, for every i each k^i -th element adds another $+1$ to the answer, and there are $\lfloor \frac{n}{k^i} \rfloor$ such elements.

The final answer is

$$\lfloor \frac{n}{k} \rfloor + \lfloor \frac{n}{k^2} \rfloor + \dots + \lfloor \frac{n}{k^i} \rfloor + \dots$$

This result is also known as [Legendre's formula](#). The sum is of course finite, since only approximately the first $\log_k n$ elements are not zeros. Thus, the runtime of this algorithm is $O(\log_k n)$.

Implementation

```
int fact_pow (int n, int k) {
    int res = 0;
    while (n) {
        n /= k;
        res += n;
    }
}
```